From this proof to a product

The reusable product is a shipment-document workflow: capture a pickup acknowledgement, retain a version and retrieve the right evidence. The browser proof demonstrates that connection. Native Android, a customer ERP and production operations still need implementation and validation.

Keep the core; replace the edges. Retain typed shipment/document/assessment contracts, explicit receipt states, idempotent operations and the evidence workspace. Put the PDF viewer, signature placement and flattened export behind a document adapter. Implement the production native Android adapter against the selected Nutrient Android SDK and entitlement. Verify it on the actual rugged tablets, including touch targets, shared-device reset, file reopening, failure recovery and any required offline queue. The web proof does not verify native Android behavior.

Make integration configurable. Define ERP interfaces for shipment lookup, source-document retrieval and a durable capture receipt. Use a storage interface for document versions, authorized retrieval, retention and export. Map stable customer shipment/invoice/PO identifiers explicitly; validate conflicts rather than assuming a match. Configure each BOL template's pickup region and acknowledgement wording. Keep receiving evidence separate. Agree the retailer's acceptance and dispute rules before labelling a packet sufficient.

Make it safe to operate. Add the customer's identity provider, role and plant permissions, shared-tablet session controls and authorization on every document route. Preserve originals and version references. Record who captured, received, accessed or assessed a record, with timestamps and operation IDs. Hashes aid version checks; they do not authenticate a signer or make a PDF certificate-backed. Package deployment, migrations, encrypted storage, backups and restore checks, monitoring, support ownership and an agreed incident/retention policy. Pilot in one plant before expanding.

Cost basis - planning assumptions, not a quote. Assume a 40-hour engineer-week and a blended **$150/hour** for engineering/QA. Hardening the reusable core, native adapter and deployment: **320-560 hours ($48k-84k)**. A new customer's ERP mapping, templates and pilot: **80-160 hours ($12k-24k)**. Combined initial allowance: **$60k-108k**, subject to architecture and acceptance requirements. Changes such as complex offline conflict handling or certificate signing need separate scope.

Budget **4-12 support hours/month ($600-1,800)** and a provisional **$100-300/month infrastructure reserve** for a modest deployment; validate storage, retention, resilience and traffic before quoting. At the customer-reported 130,000 BOLs/year, assumed 0.5-2 MB files mean roughly **65-260 GB/year per retained copy**, before backups. Nutrient Android/Web/server entitlements, device/MDM costs and any external service or premium SLA are **unquoted and excluded**. Confirm licensing with Nutrient; do not infer production rights or price from a working trial. Validate actual implementation effort and customer demand in the pilot before selling it as ready made.